Task Design Document:

FR4 & CatFR4

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Version	Author	Description	Date
1.0.0	Maciej Swat	First version	16MAR2016
1.0.1	Stas Busygin	Edits	01APR2016
1.0.2	Dan Rizzuto	Combined FR4 and CatFR4 task design	01JUL2016

Overview

The FR4 task implements open loop brain stimulation during a memory task to test the hypothesis that direct electrical stimulation (DES) of the brain during poor memory states results in better memory than a control condition with no brain stimulation. This work builds on three prior experiments: FR1, FR2 and FR3.

In FR1 we recorded brain activity from multiple implanted electrodes during the encoding and retrieval of word lists and identified neural correlates of good and poor memory encoding states (henceforth biomarkers). We sought to predict whether a given item would be recalled based on the spectral decomposition of the time-series of bipolar neural recordings from all N channels recorded. We thus fit a L2-regularized logistic regression model to the item level data in which the LHS variable was the recalled/non-recalled status of an item and the RHS variables were the wavelet power in each of 8 frequency bands and across all N channels (henceforth classifier). The model was fit to data from sessions 1 through M-1 and validated on session M, with the penalty parameter determined based on the distribution of best fitting parameters across all patients who performed the experiment excluding the patient whose data were being fit. The model was able to reliably predict recall based on the spectral decomposition of the encoding period data, as described by Ezzyat et al (submitted).

In FR2 we used open loop brain stimulation to assess whether DES would improve or impair memory for the items immediately following the cessation of stimulation. Although we found that DES frequently resulted in impaired memory there were some cases of memory improvement. Using the logistic regression model described above we found that patients for whom DES resulted in improved classifier output (i.e., predicting better memory) exhibited better memory for the post-stimulated items. In contrast, those patients for whom DES resulted in reduced classifier output exhibited worse memory for the post-stimulated item. We further hypothesized that DES is more likely to improve memory when it arrives during poor memory states than when it arrives during good memory states. We thus partitioned items into terciles based on the state of the classifier and found that DES resulted in improved memory when it arrived during the lowest tercile; no such beneficial effects were observed when DES arrived during higher terciles of classifier output.

In FR3 we apply DES in a closed-loop mode where we use 1366 ms worth of EEG data following word onset to evaluate L2 logistic regression classifier (as described above) and apply DES when classifier output is below threshold. Since evaluating classifier requires approximately 120ms the stimulation will take place at approximately 1486 ms since word onset. The threshold is determined to be the median of the distribution of the classifier output for each individual subject (see FR3 Task Design document for details). We stimulate in the low biomarker states that correspond to poor memory and assume that applying DES in those states has higher probability of improving memory state that applying DES in good memory states.

To verify the validity of this assumption and to establish control group for statistical analysis we propose a new experiment - FR4, where we will apply DES to every other word (excluding first

and last item from the list) irrespective of the biomarker indication (see Methods section for details). The stimulation parameters will be determined prior to FR4 session using PS data.

Methods

Except as noted, the design of the behavioral task and the procedure for it are identical to FR3. The reader is referred to that design document for further details. Subjects will perform delayed free recall on 26 study-test lists, the first of these being practice (all data from all lists, including the practice list, will be recorded). Stimulation is not delivered to the brain during lists 1-4 to match the design of the FR3 experiment.

Following the first 4 lists, the remaining 22 lists will be split into 3 groups: 8 no-stim lists (NoStim), 7 lists where we will stimulate odd numbered items (StimOdd) and 7 lists where we stimulate even numbered items (StimEven). An item is considered stimulated if stimulation occurred during the end of the previous word and beginning of the previous ISI. We start numbering of items from 0 and exclude 0th and last item from being stimulated. These 22 lists will be pseudo-randomly interleaved to avoid any systematic bias associated with fixed order of the lists.

Closed-loop Procedure

During the item presentation phase of the StimEven or StimOdd lists in FR4 we will apply DES after 1366ms+120ms=1486ms since word onset of even or odd numbered items respectively. The stimulation duration is set to 500ms .

Stimulation Artifacts. The stimulation will be delivered to every other word which will ensure that stimulation artifacts subside before the next stimulation event

Selection of Open-Loop Stimulation Parameters

To match the design of the FR3 experiment the duration of stimulation will be set to 500 ms. Other parameters of the stimulation such as location, amplitude, pattern frequency or train frequency will be set individually for each patient.

Categorized Free Recall

The CatFR4 task is identical to the FR4 task described in this document with the only exception being that this task will use the word pool from the CatFR1/2 tasks.

References

- Hastie. T., Tibishirani, R., & Friedman, J. (2001). The elements of statistical learning: data mining, inference and prediction. *New York: Springer*.